Technology
Arts Sciences
TH Köln

Faculty 07
Institute of Computer and Communication Technology

Basics on Systems and Networks

Part 3: Lab on Digital Modulation

Version 1.0

June 10, 2026

1 Introduction

In conjunction with the in-presence session of the course Basics on Systems and Networks, participants will have the opportunity to refresh or expand their knowledge of digital modulation schemes in this lab exercise. Since the presence time is quite limited the tasks shall be performed self-conducted at home and the results be presented during a 10 minute time slot on June 26, 2026.

Learning outcomes

- Understand linear modulation schemes (PAM, PSK, QAM)
- Constellation diagrams with Gray coding
- Coherent reception and optimal detection
- Bit error rate (BER) for AWGN channel
- Performance comparison of different schemes

2 Jupyter Notebook to simulate and compare some basic digital modulation schemes.

This task deals with some basic digital modulation schemes and their properties. To solve this task please perform the following steps:

2.1 Work Instructions:

2.1.1 At home

- Answer all questions in the Jupyter Notebook either in the notebook itself or on a sheet of paper
- Go through the simulation cells, try to understand the code and interprete the results
- Fill the code gaps marked by question marks (??)

2.1.2 At presence time

- Get a timeslot during the presence time and present and explain your results
- You must be able to run all simulations on your laptop

Note: *Copying answers or using AI for solving the problems is not permitted and will not be successful since you will be tested during your presentation in the presence time. Questions will vary!*

If you have not installed Python or Jupyter notebook before we recommend installing the anaconda distribution under `https://www.anaconda.com/download` .

3 Timetable

The following table shows the schedule for the presentation of your lab results.

Date	Task
26.6.26 between 1 pm and 4 pm	Jupyter Notebook discussion

It is mandatory for the students to be able to present the running simulations files, to answer the questions, and to explain the results to achieve the ULP.